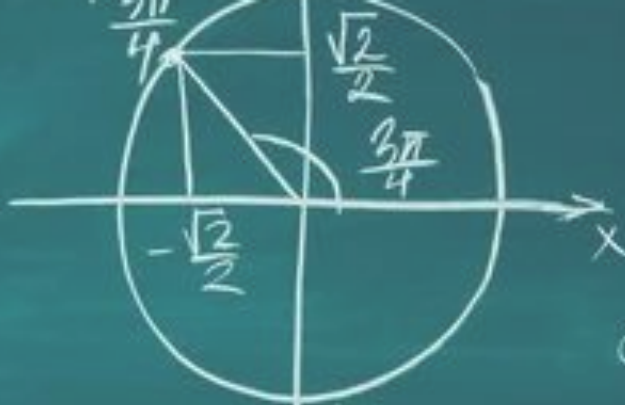
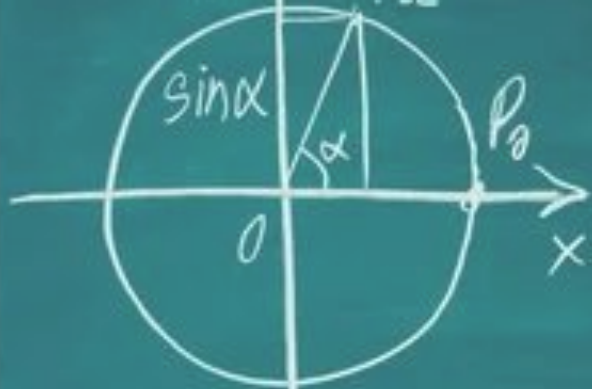




$$\operatorname{tg} \frac{3\pi}{4} = -1$$

$$\sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}, \quad \cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}, \quad \operatorname{tg} \frac{3\pi}{4} = -1$$

$$D(\sin) = D(\cos) = 0, \quad E(\sin) = E(\cos) = [1; 1]$$

First year Aerospace Engineering

Mathematics Recap - Trigonometry

Marcos Carrera Valle

Amplitude, Period, Frequency

Using the unit circle, solve

1) $\sin(0)$ 7) $\cos(0)$

2) $\sin(\frac{\pi}{6})$ 8) $\cos(\frac{\pi}{2})$

3) $\sin(\frac{\pi}{4})$ 9) $\cos(\frac{\pi}{6})$

4) $\sin(\frac{\pi}{3})$ 10) $\cos(\frac{2\pi}{3})$

5) $\sin(\frac{\pi}{2})$ 11) $\cos(\frac{5\pi}{6})$

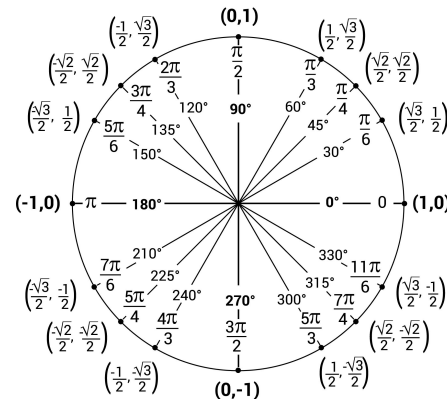
6) $\sin(\pi)$ 12) $\cos(\pi)$

13) $\tan(0)$

14) $\tan(\frac{\pi}{6})$

15) $\tan(\frac{\pi}{4})$

16) $\tan(\frac{\pi}{3})$



Amplitude, Period, Frequency - 6 points

Without sketching, for the function $f(x) = 3\cos(2x - \frac{\pi}{4}) - 3$, $0 < x < \pi$

- 1) What is the amplitude - 1 point
- 2) What is the period - 1 point
- 3) What is the frequency - 1 point
- 4) What is the mean value - 1 point
- 5) What is the value of $f(\frac{\pi}{8})$
- 6) What is the phase at $f(x) = 0$

Amplitude, Period, Frequency

Convert the following angles from degrees to radians

- 1) 0°
- 2) 30°
- 3) 45°
- 4) 90°
- 5) 60°
- 6) 270°
- 7) 305°

Pythagorean Identities

- 1) If $\sin(\theta) = \frac{4}{5}$, find $\cos(\theta)$ using the pythagorean theorem
- 2) **Without using your calculator**, find $\cos\left(\frac{7\pi}{12}\right)$

Double angle formulae

1) Solve $\int (1)/(\sin(2x)\cos(2x)) * (2\tan(u))/(1-\tan^2(u)) du$

Answers

Slide 2:

1) 0

2) $\frac{\sqrt{1}}{2}$

3) $\frac{\sqrt{2}}{2}$

4) $\frac{\sqrt{3}}{2}$

5) 1

6) 0

7) 1

8) 0

9) $\frac{\sqrt{3}}{2}$

10) $-\frac{\sqrt{1}}{2}$

11) $-\frac{\sqrt{3}}{2}$

12) -1

13) 0

14) $\frac{1}{\sqrt{3}}$

15) 1

16) $\sqrt{3}$

Answers

Slide 3:

- 1) Amplitude = 3
- 2) Period = π
- 3) Frequency = 2
- 4) Mean value = -3
- 5) $f(\frac{\pi}{8})=0$
- 6) $\pi/8$

Answers

Slide 4:

1) 0

2) $\frac{\pi}{6}$

3) $\frac{\pi}{4}$

4) $\frac{\pi}{2}$

5) $\frac{2\pi}{3}$

6) $\frac{3\pi}{2}$

7) $\frac{17\pi}{10}$

Answers

Slide 5:

1) $\frac{3}{5}$

2) $(-\sqrt{6} + \sqrt{2}) / 4$

Double angle formulae

1) Solve $\int (1)/(\sin(2x)\cos(2x)) * (2\tan(x))/(1-\tan^2(x)) dx$

$$\frac{1}{2} \tan(2x) + C$$